YOLOv8 for Empty Parking Space Detection: Enhancing Efficient Vehicle Parking Management

[1st] Muhammad Irsyad Asyarie, Robotics and Artificial Intelligence Engineering, Universitas Airlangga, Surabaya, Indonesia (email: muhammad.syad.asyarie-2021@ftmm.unair.ac.id)
[2nd] Muhammad Izza Dwi N. A, Robotics and Artificial Intelligence Engineering, Universitas Airlangga, Surabaya, Indonesia (email: muhammad.syad.asyarie-2021@ftmm.unair.ac.id)
3rd Maharani Yassar Dewanti, Robotics and Artificial Intelligence Engineering, Universitas Airlangga, Surabaya, Indonesia (email: maharani.yassar.dewanti-2021@ftmm.unair.ac.id)

Abstract— **As urbanization continues to grow, the increasing number of vehicles on city streets highlights the urgency of smart parking guidance systems. This document addresses the need to provide real-time information to drivers about the nearest parking lots and the number of available spaces. Leveraging a vision-based approach, the proposed method uses YOLOv8, an advanced target detection algorithm, to detect empty parking spaces using images captured by surveillance cameras. The method involves using YOLOv8 to analyze images from surveillance cameras, thereby providing accurate and real-time information on parking space occupancy. This system is designed based on vision, leveraging the capabilities of computer vision to improve the effectiveness of parking guidance. The algorithm's ability to detect both parking spaces and vehicles contributes to an overall understanding of parking lot occupancy. The integration of YOLOv8 in the proposed system ensures robust detection performance, considering the speed and accuracy of the algorithm. Surveillance cameras act as data collection tools, collecting information about parking conditions and the presence of vehicles. The YOLOv8-based approach facilitates the creation of an intelligent and responsive parking guidance system, capable of meeting the dynamic needs of urban parking environments.**

Index Terms—**Parking lot, vehicle detection, yolo, parking management system.**

I. Introduction

The growing urban populations and corresponding increases in vehicle ownership have exacerbated the challenge of effective parking management. Searching for empty parking spaces in crowded urban areas not only contributes to traffic congestion but also leads to increased pollution and wasted time. To solve this urgent problem, the introduction of vision-based parking management systems has emerged as a promising solution. This paper presents a state-of-the-art approach using YOLOv8, an advanced deep learning model, to detect empty parking spaces. Our approach aims to improve parking management efficiency by providing accurate, real-time information on parking availability, thereby facilitating traffic flow.

The exponential growth of urbanization and the corresponding increase in the number of vehicles has posed a significant challenge to the effective management of parking lots. Traditional parking management methods are often manual, time-consuming and error-prone, leading to increased traffic congestion and driver frustration. The need for smart parking solutions is obvious, and one particular aspect that needs attention is the ability to accurately, real-time detect empty parking spaces.

Current parking management systems often lack the accuracy and efficiency needed to quickly identify vacant parking spaces, leading to suboptimal use of available parking resources. This inefficiency worsens traffic congestion, increases carbon emissions, and reduces the overall quality of urban life. To address these challenges, the research focuses on leveraging YOLOv8, an advanced object detection model, to improve the accuracy and speed of empty parking space

This work was supported by Airlangga University. Faculty of Advance Technology and Multidiscipline, Department of Robotics and Artificial intelligence Engineering, Surabaya, Indonesia.

M. I. Asyarie is with Airlangga University, Faculty of Advance Technology and Multidiscipline, Department of Robotics and Artificial intelligence Engineering, Surabaya, Indonesia (e-mail: muhammad.syad.asyarie-2021@ftmm.unair.ac.id).

M. I. D, N, Ajiwardoyo is with Airlangga University, Faculty of Advance Technology and Multidiscipline, Department of Robotics and Artificial intelligence Engineering, Surabaya, Indonesia (e-mail: muhammad.izza.dwi-2021@ftmm.unair.ac.id).

M. Y, Dewanti is with Airlangga University, Faculty of Advance Technology and Multidiscipline, Department of Robotics and Artificial intelligence Engineering, Surabaya, Indonesia (e-mail: maharani.yassar.dewanti-2021@ftmm.unair.ac.id).

detection.

The main challenge lies in developing a system that seamlessly integrates YOLOv8 into the existing parking management infrastructure, ensuring that the system reliably identifies empty parking spaces in a variety of environments. Different fields and lighting conditions. Addressing this challenge will help create a more responsive and automated parking management system, optimizing parking space utilization and easing pressure on urban transportation networks.

The purpose of this research is to explore the capabilities of YOLOv8 in detecting empty parking spaces, addressing the limitations of current parking management systems and providing a basis for the development of intelligent solutions. Real-time intelligence to improve parking. in urban areas. setting.

II. Related Works

The proposed approach in this study focuses on developing a data-driven Empty Parking Space Detection (EPSD) system using YOLO (You Only Look Once) for detecting vacant parking spaces in large parking areas. The methodology is centered around the generation of assessments from various data sources, employing leading machine learning algorithms as baseline models. The rationale behind this approach is that incorporating domain knowledge in the context of parking can enhance the performance of the developed model. The literature review in this section is subdivided into the ground truth data used for validation in parking studies, supplementary data utilized for feature engineering independent of ground truth data, and popular parking prediction approaches utilizing YOLO. Additionally, the review encompasses the use of behavior change detection models in EPSD systems, with a specific focus on vacant space detection in expansive parking areas..

To insert images in *Word,* position the cursor at the insertion point and either use Insert | Picture | From File or copy the image to the Windows clipboard and then Edit | Paste Special | Picture (with "float over text" unchecked).

IEEE will do the final formatting of your paper. If your paper is intended for a conference, please observe the conference page limits.

A. Ground Truth Data Used for Validation of Parking Study

The proposed approach in this study focuses on developing a data-driven Empty Parking Space Detection (EPSD) system using the YOLO (You Only Look Once) methodology for identifying vacant parking spaces in expansive parking areas. The methodology is centered around the generation of assessments from various data sources, leveraging YOLO as the primary algorithm for real-time object detection in large parking lots. The rationale behind this approach is that incorporating advanced machine learning techniques, such as YOLO, can significantly enhance the accuracy and efficiency of parking space detection.

Similar to the existing state-of-the-art on-street parking availability models, the data sources for EPSD can be classified into two main categories: data used exclusively for feature engineering and ground truth data primarily used for training, testing, and validating the system. Different types of ground truth data are explored for the validation of the empty parking space prediction model. Some studies have utilized parking sensors [8], [9], [11], [12], [13], [14], [15], while others have employed innovative approaches such as YOLO-based image and video processing from cameras mounted on moving vehicles [2], [20]. Additionally, the study proposes the integration of YOLO for real-time detection and classification of parking spaces in large parking areas, utilizing both ground truth data and automatically collected sparse parking events data.

An essential aspect of this research is addressing the potential limitations of traditional ground truth collection methods, such as manual observations, and exploring YOLO as a means to supplement and support manual ground truth collection. By incorporating YOLO, which is capable of efficiently processing images and videos, the study aims to reduce the frequency of manual observations required in practice. The focus is on testing the effectiveness of YOLO-based methods in comparison to traditional ground truth data collection, emphasizing the potential benefits of automating and enhancing the accuracy of the empty parking space detection system in large parking lots.

B. Popular Parking Prediction Models

Object detection is a fundamental task in computer vision, involving the identification of specific features within images [5]. Teoh and Bräunl [6] employ a multi-size symmetric search window to extract symmetric regions, validated by the Adaboost classifier. Teutsch and Kruger [7] utilize sliding windows to locate candidate vehicles from airborne cameras, employing an AdaBoost classifier based on motion features for verification.

In recent years, deep learning-based target detection methods have gained prominence. K. Simonyan and A. Zisserman [8] proposed a very deep CNN convolutional network to construct a VGG network for object classification. RCNN [9] (Region Proposal-Convolutional Neural Network) combines Region Proposal and Convolutional Networks for target detection. S. Ren et al. [10] introduced Faster R-CNN, more effective than RCNN, abandoning selective search and introducing the RPN network for accelerated detection. YOLO [11] integrates target discrimination and recognition, enhancing detection speed. YOLO v3 [12] utilizes multi-scale features, achieving 57.9% mAP effectiveness in 51 ms on VOC datasets, ensuring accuracy and detection rate in target detection.

In parking lot management, beyond vehicle detection, identifying vacant parking spaces poses a challenge. M. Ahrnbom et al. [13] extract features and employ SVM-based classifiers for parking lot occupancy classification. Giuseppe Amato et al. [14] use deep CNN to train detectors for parking space detection based on LBP features. Tom Thomas et al. [15] construct a binary classifier convolutional neural network to judge parking lot occupancy. Cheng-Fang Peng [16] extracts three features for each parking space, training a deep neural network to determine occupancy status.

However, the existing system [14] relies on a manually created mask for identifying parking spaces. In this study, we propose using YOLO v8 instead of YOLO v3, aiming to improve the efficiency and accuracy of parking space detection in diverse parking lots. The upgraded YOLO version is anticipated to enhance real-time detection performance and contribute to automating

III. Methodology

In the dynamic realm of computer vision and object detection, YOLOv8 emerges as a cutting-edge solution, representing the latest evolution in the renowned YOLO (You Only Look Once) series. Developed to address the ever-growing demands for accuracy, speed, and versatility, YOLOv8 stands at the forefront of real-time object detection methodologies.

Building upon the success of its predecessors, YOLOv8 incorporates advancements in deep learning, leveraging a state-of-the-art convolutional neural network architecture. This latest iteration not only maintains the hallmark efficiency of YOLO but also introduces enhancements that propel it to new heights in terms of detection precision and adaptability.

As we delve into the intricacies of YOLOv8, its capacity to seamlessly integrate target discrimination and recognition, coupled with multi-scale features, becomes apparent. These features collectively contribute to a robust and efficient system, making YOLOv8 a formidable tool in a diverse range of applications, from security and surveillance to autonomous vehicles and beyond.

A. Model Structure: Deep Dive

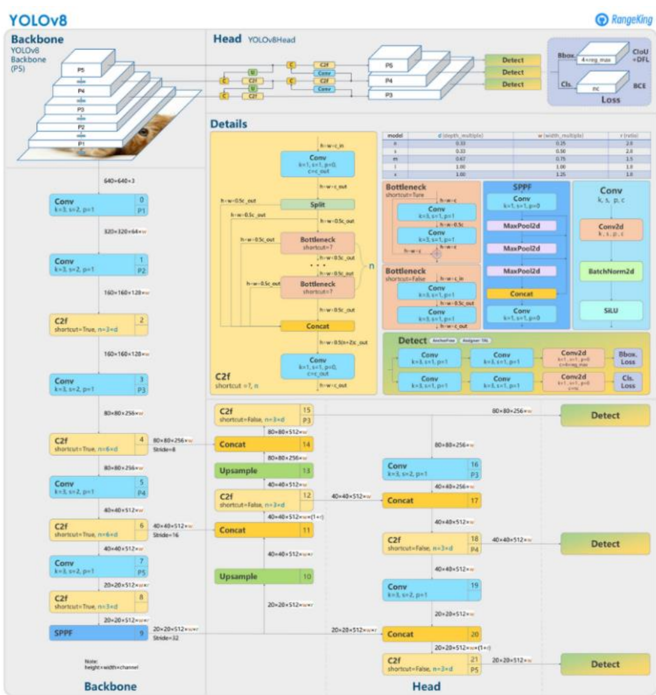

Figure 3.1 YOLOv8 Architecture

YOLOv8 is an anchor-free model. This means it predicts directly the center of an object instead of the offset from a known anchor box.

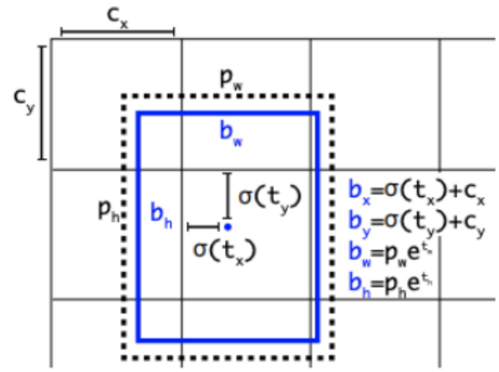

Figure 3.2 Visualization of an anchor box in YOLOv8

Anchor boxes were a notoriously tricky part of earlier YOLO models, since they may represent the distribution of the target benchmark's boxes but not the distribution of the custom dataset.

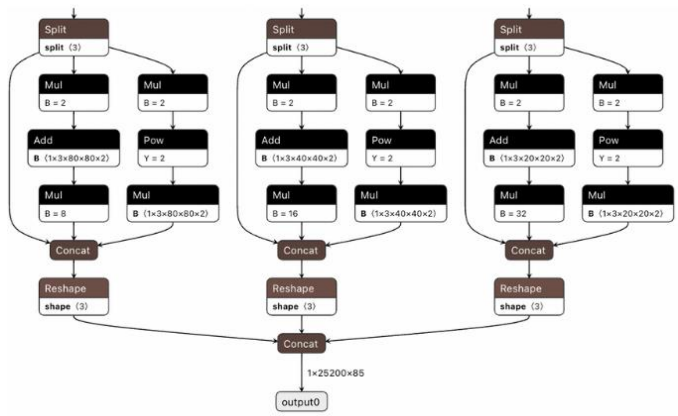

Figure 3.3 The detection head of YOLOv8

Anchor free detection reduces the number of box predictions, which speeds up Non-Maximum Suppression (NMS), a complicated post processing step that sifts through candidate detections after inference.

Anchor free detection reduces the number of box predictions, which speeds up Non-Maximum Suppression (NMS), a complicated post processing step that sifts through candidate detections after inference.

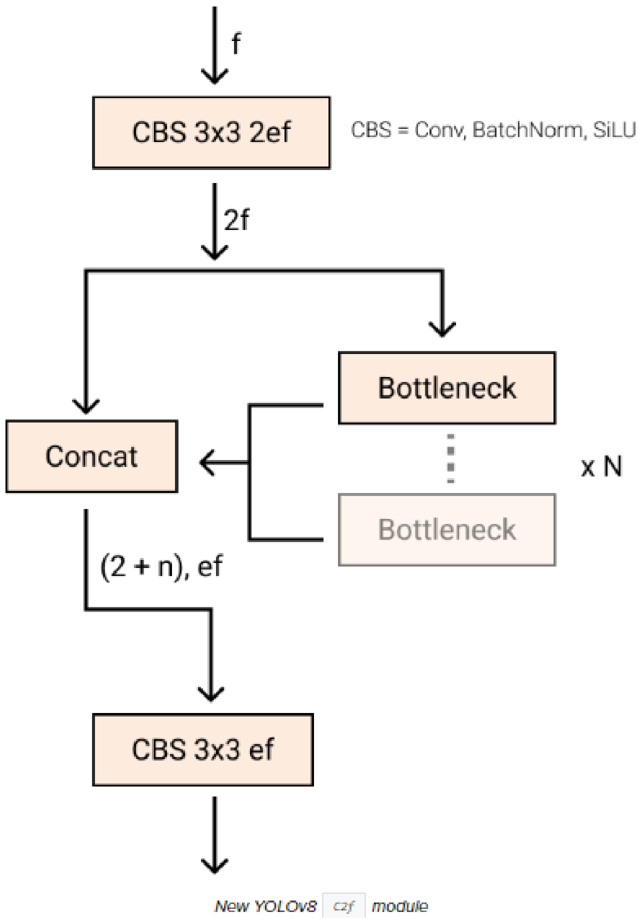

Figure 3.4 The Module of YOLOv8

The Bottleneck is the same as in YOLOv5 but the first conv's kernel size was changed from 1x1 to 3x3. From this information, we can see that YOLOv8 is starting to revert to the ResNet block defined in 2015. In the neck, features are concatenated directly without forcing the same channel dimensions. This reduces the parameters count and the overall size of the tensors.

Deep learning research tends to focus on model architecture, but the training routine in YOLOv5 and YOLOv8 is an essential part of their success. YOLOv8 augments images during training online. At each epoch, the model sees a slightly different variation of the images it has been provided. One of those augmentations is called mosaic augmentation. This involves stitching four images together, forcing the model to learn objects in new locations, in partial occlusion, and against different surrounding pixels. However, this augmentation is empirically shown to degrade performance if performed through the whole training routine. It is advantageous to turn it off for the last ten training epochs. This sort of change is exemplary of the careful attention YOLO modeling has been given in overtime in the YOLOv5 repo and in the YOLOv8 research.

IV. Results Analysis

The implemented program utilizing YOLOv8 for parking space detection has been integrated with functionality that allows the anchor boxes, representing parking spaces, to change color to red when occupied by a vehicle. This feature serves to provide a visual signal to users and facilitates the identification of occupied parking spaces.

When YOLOv8 detects the presence of a vehicle in a parking space, the anchor box encompassing that area will automatically change color to red. This color change is dynamic and provides a direct indication to users or system operators that the parking space is occupied.

Furthermore, as part of the automated response, the system will assign a parking space number to the user for vacant parking spaces. This parking space number can be provided through a user interface or integrated into a larger parking system. This process enhances parking management efficiency by delivering more detailed and immediate information to users, allowing them to quickly identify and navigate to available parking spaces. Following are the video images taken;

Figure 4.1 Visualization of an anchor box in parking area

Figure 4.2 The system is able to distinguish

A deeper dive into the model structure reveals important features of YOLOv8. The implementation of ResNet blocks in bottlenecks, along with changes in the size of convolutional kernels, shows a thoughtful evolution in design aimed at optimizing model performance. Combining features at the neck, without forcing the same channel dimensions, further reduces the number of parameters and overall tensor size, thereby increasing the computational efficiency of the model.

YOLOv8's success is due not only to its architecture but also to its carefully designed training routines. The introduction of mosaic augmentation, which combines four images into one during training, forces the model to learn objects in different locations, in partial occlusion, and at different surrounding pixels. This dynamic training approach contributes to the model's adaptability and robustness in handling diverse parking lot environments.

V. SUMMARY

In conclusion, this article addresses the growing challenges of urbanization and increasing vehicle ownership by proposing a smart parking guidance system using YOLOv8, an advanced target detection algorithm. The system aims to provide real-time information to drivers about the nearest parking lots and available spaces. Leveraging a vision-based approach with surveillance cameras, YOLOv8 detects empty parking spaces and vehicles, contributing to an improved understanding of parking lot occupancy. The integration of YOLOv8 ensures robust detection performance, and the proposed system is designed to be intelligent and responsive, meeting the dynamic needs of urban parking environments. The article also discusses the challenges of traditional parking management methods and emphasizes the importance of accurately and in real-time detecting empty parking spaces. The methodology involves using YOLOv8 for Empty Parking Space Detection (EPSD), leveraging various data sources for validation and training. The study explores the limitations of traditional ground truth collection methods and highlights the potential benefits of automating and enhancing the accuracy of empty parking space detection using YOLOv8. The article delves into popular parking prediction models and introduces YOLOv8 as a cutting-edge solution, providing a deep dive into its architecture and training routines. Finally, the results analysis reveals the successful integration of YOLOv8, enabling dynamic color changes in anchor boxes and automated assignment of parking space numbers for vacant spaces. The article concludes with a summary of the achieved outcomes and the potential for optimizing parking space utilization in urban environments.

REFERENCES

[1] A. Alekhina, P. Korolev, and M. Dorrer, "Application of convolutional networks of the YOLO architecture for calculating free parking spaces in an arbitrary parking lot," E3S Web of Conferences, vol. 458, no. 03024, 2023. doi:10.1051/e3sconf/202345803024.

[2] Pawar, A. Pawar, A. Pawar, G. Pawar, and A. Chaudhari, "An Elaborative Study of Smart Parking Systems," International Journal of Engineering Research & Technology (IJERT), vol. 10, no. 10, October 2021. [Online]. Available: http://www.ijert.org. [Accessed: Dec. 17, 2023]

[3] A. R. V. N. Sai, "Car Parking Occupancy Detection using YOLOv3," Master of Engineering thesis, School of Engineering and Technology, Asian Institute of Technology, Thailand, May 2019

[4] B. Li, "Detection of Parking Spaces in Open Environments with Low Light and Severe Weather," in ISEMSS 2022, ASSEHR 687, G. Ali et al., Eds., pp. 3011–3020, 2022, doi: 10.2991/978-2-494069-31-2_354.

[5] D.-L. Nguyen, X.-T. Vo, A. Priadana, and K.-H. Jo, "YOLO5PKLot: A Parking Lot Detection Network Based on Improved YOLOv5 for Smart Parking Management System," in Dept. of Electrical, Electronic and Computer Engineering, University of Ulsan, Ulsan 44610, South Korea.

[6] E. Ektrada, L. Hakim, and S. P. Kristanto, "Sistem Tracking dan Counting Kendaraan Berbasis YOLO untuk Pemetaan Slot Parkir Kendaraan," SESSION (Software Development, Digital Business Intelligence, and Computer Engineering), vol. 01, no. 02, pp. 55-60, Mar. 2023. (yang ini bukan bhs inggris ran)

[7] I. H. Jung, J.-M. Lee, and K. Hwang, "Advanced Smart Parking Management System Development Using AI," Journal of System and Management Sciences, vol. 12, no. 1, pp. 53-62, 2022. DOI:10.33168/JSMS.2022.0105.

[8] I. H. Jung, J.-M. Lee, and K. Hwang, "Advanced Smart Parking Management System Development Using AI," Journal of System and Management Sciences, vol. 12, no. 1, pp. 53-62, 2022. DOI: 10.33168/JSMS.2022.0105.

[9] O. Tran Thi Kim, N. H. Tran, C. Pham, T. LeAnh, M. T. Thai, and C. S. Hong, "Parking Assignment: Minimizing Parking Expenses and Balancing Parking Demand Among Multiple Parking Lots," in IEEE Transactions on Automation Science and Engineering, vol. 17, no. 3, pp. 1320-1330, July 2020. doi: 10.1109/TASE.2019.2948200.

[10] R. Grbić and B. Koch, "Automatic vision-based parking slot detection and occupancy classification," Expert Systems With Applications, vol. 225, no. 120147, Apr. 2023. [Online]. Available: https://doi.org/10.1016/j.eswa.2023.120147.

[11] S. Gomari, R. Domakuntla, C. Knoth, and C. Antoniou, "Development of a Data-Driven On-Street Parking Information System Using Enhanced Parking Features," in IEEE Open Journal of Intelligent Transportation Systems, vol. 4, no. 1, pp. 30-47, Jan. 2023. doi: 10.1109/OJITS.2023.3235898.

[12] S. Liu, Y. Liu, J. Xiong, and T. Zhang, "Parking Spot Detection with YOLOv5," KTH Royal Institute o

[13] f Technology, Stockholm, Sweden, 2023. [Online]. Available: [URL of the article]. [Accessed: 17-Dec-2023].

[14] S.-E. Shih and W.-H. Tsai, "A Convenient Vision-Based System for Automatic Detection of Parking Spaces in Indoor Parking Lots Using Wide-Angle Cameras," in IEEE

[15] Transactions on Vehicular Technology, vol. 63, no. 6, pp. 2521-2532, July 2014. doi: 10.1109/TVT.2013.2297331.

[16] S. Gomari, R. Domakuntla, C. Knoth, and C. Antoniou, "Development of a Data-Driven On-Street Parking Information System Using Enhanced Parking Features," in IEEE Open Journal of Intelligent Transportation Systems

[16] vol. 4, no. 1, pp. 30-47, Jan. 2023. doi: 10.1109/OJITS.2023.3235898.

[17] V. Paidi, H. Fleyeh, J. Håkansson, and R. G. Nyberg, "Tracking Vehicle Cruising in an Open Parking Lot Using Deep Learning and Kalman Filter," Journal of Advanced Transportation, vol. 2021, Article ID 1812647, Aug. 2021. doi:10.1155/2021/1812647. Available: https://doi.org/10.1155/2021/1812647.

[18] X. Ding and R. Yang, "Vehicle and Parking Space Detection Based on Improved YOLO Network Model," in J. Phys.: Conf. Ser., vol. 1325, no. 012084, 2019, doi:10.1088/1742-6596/1325/1/012084.